

SAM2Depth: Video Depth Estimation for Video Segmentation

Tymoteusz Kwieciński

Sara Lazarusic

Lovro Nidogon

Mariia Shpir

Department of Computer Science, ETH Zürich

Abstract

Frame-by-frame monocular depth estimation on video often produces temporal flicker and unstable object boundaries. We propose to reduce this effect by using SAM3 to provide object-centric and trackable scene units. We use SAM3 decoder queries (tracked across frames) as discrete tokens inside an iDisc-style depth decoder and enforce instance-level temporal consistency. We evaluate standard depth accuracy and temporal stability over tracked instances.

1. Motivation

Monocular depth estimation predicts per-pixel depth from RGB images, but when applied to videos it often shows *temporal flicker*: the same object changes depth across nearby frames [5]. This is problematic for downstream tasks such as video segmentation and tracking, which assume temporally consistent geometry.

We build on iDisc, which improves depth estimation by compressing dense image features into a small set of discrete internal representations and decoding them back to dense features before predicting depth [3]. Separately, query-based segmentation models decode masks using a small set of learned queries and cross-attention; the attention weights can be interpreted as soft masks [2].

SAM3 provides segmentation and tracking in video with persistent instance identities [1]. Our hypothesis is that using these tracked, object-centric query representations as discrete units for depth decoding will reduce flicker and improve boundary consistency.

2. Method

Base model (iDisc). iDisc follows an encoder–decoder design and introduces an internal discretization bottleneck: dense pixel features are summarized into a small set of discrete representations and then decoded back to dense features for depth prediction [3].

Main integration (SAM3 queries as discrete tokens).

We replace (or augment) the discrete representations in iDisc with object-centric tokens derived from SAM3, which are consistent across frames due to tracking [1].

Pipeline (per video clip).

1. **SAM3 (frozen, cached):** run SAM3 on the clip and store (i) tracked instance IDs, (ii) masks, and (iii) the corresponding decoder query embeddings when accessible.
2. **Depth backbone:** encode each frame to obtain multi-scale pixel features.
3. **Token-to-pixel fusion:** apply cross-attention where pixel features query the SAM3 tokens (keys/values), mirroring iDisc’s discrete-to-continuous decoding step.
4. **Depth head:** predict a dense depth map (multi-scale fusion as in iDisc).

Temporal consistency loss (instance-level). For each tracked instance k , we compute the median predicted depth inside its mask at each frame t and penalize its temporal variance across the clip (applied only to sufficiently long, confident tracks).

Alternative (mask-based tokens). If SAM3 query embeddings are not accessible, we build one token per instance by masked pooling of depth-backbone features (mask resized to feature resolution). The downstream fusion and depth head remain unchanged, but performance may drop due to mask errors and resolution-dependent leakage.

3. Evaluation

Datasets. We use NYU Depth V2 (indoor) and KITTI (driving) as primary supervised benchmarks. If time permits, we add one additional outdoor dataset (e.g., DDAD) to test generalization.

Baselines / variants.

1. iDisc (no SAM3) [3];
2. iDisc + SAM3 query tokens (no temporal loss);
3. full method (SAM3 query tokens + temporal loss);
4. alternative: mask-based tokens.

Depth accuracy metrics. We report AbsRel, RMSE, and $\delta < 1.25$ (and otherwise follow the iDisc evaluation protocol) [3].

Temporal stability metrics. To assess temporal consistency and reduced flicker, we employ both dense and instance-aware metrics:

- *Dense consistency* (τ_5): Following VeloDepth [4], we measure pixel-wise temporal stability across frames. We warp the depth from $t - 1 \rightarrow t$ using optical flow, apply ego-motion correction, and report the percentage of inlier pixels (difference $< 5\%$) after masking out occlusions via forward-backward flow consistency.
- *Tracked instances stability*: For each tracked instance, we compute the τ_5 metric specifically within its mask to measure object-level structural consistency. Additionally, we report the variance of the median depth inside the mask across time to capture macroscopic object stability.

We additionally provide qualitative videos highlighting object-wise stability.

4. Risks and Alternatives

R1: Access to SAM3 query embeddings. SAM3 query embeddings may not be exposed by the public interface or may be hard to match to persistent instance IDs. We will inspect SAM3 inference outputs and, if needed, modify the inference wrapper to log per-instance query embeddings and cache them offline. *Alternative*: if queries are not accessible, we construct one token per tracked instance by masked pooling of depth-backbone features (mask resized to feature resolution) and keep the same downstream cross-attention decoder.

R2: Tracking noise (ID switches, missed instances). Tracking errors can enforce incorrect temporal constraints. We apply the temporal loss only to confident tracks that persist for at least $T_{\min}$ frames and ignore short or unstable tracks. *Alternative*: if tracking remains unreliable, we disable the temporal loss and report results without temporal regularization.

R3: Compute cost. Running SAM3 during training can be slow. We freeze SAM3 and precompute/cache masks, IDs, and tokens for all clips. *Alternative*: shorten clip length and limit the number of instances per frame.

5. Roadmap

- **Weeks 1–2**: reproduce iDisc baseline training/evaluation on NYU Depth V2 and KITTI.
- **Week 3**: run SAM3 on clips; implement caching (masks, IDs, queries if available).
- **Weeks 4–5**: implement token-to-pixel cross-attention fusion; train without temporal loss.

- **Week 6**: add instance-level temporal loss; tune loss weights and track filtering.
- **Weeks 7–8**: ablations, qualitative videos, final report with failure cases.

6. Expected Outcome

We expect that using SAM3-tracked object queries as discrete tokens will reduce temporal flicker in video depth estimation compared to frame-by-frame depth prediction. In particular, we expect lower instance-level depth variance over time for tracked objects and sharper depth boundaries aligned with object masks. We also expect to preserve (or slightly improve) standard depth accuracy metrics relative to the iDisc baseline when trained under the same supervision.

References

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